

MMAE 545 - Project 1: Minimum-Cost Plastic Bracket Design

Illinois Institute of Technology - Spring 2026

1. Introduction

The objective of this project is to design a plastic support bracket minimizing material cost under two structural specifications: (1) a factor of safety of at least 2.0 with respect to von Mises stress, and (2) a maximum y-direction deflection no greater than 0.015 in along the bottom edge, both under a 150 lbf uniformly distributed load. The bracket fits within an 8 in $\times$ 15 in envelope with constant z-direction thickness and vertical centerline symmetry. A multi-stage topology and parametric optimization workflow was used to obtain a constraint-saturated minimum-cost design, verified by linear static FEA in ANSYS Workbench 2025 R2 under the required 2D plane stress formulation.

2. Modeling and Optimization Procedure

2.1 Material Selection

Six candidates were evaluated using $M = E / (\rho \times \text{price})$, where E is elastic modulus (GPa), ρ is density (kg/m³), and price is cost (\$/kg). This metric identifies which material delivers the most stiffness per dollar spent, the governing tradeoff in a deflection-limited cost problem. Polypropylene ranked highest at $M = 9.88 \times 10^{-4}$ due to its favorable combination of low density (900 kg/m³), adequate modulus (1.2 GPa), and low price (\$1.35/kg). Other candidates were eliminated by high unit cost (Polycarbonate, \$3.65/kg) or insufficient stiffness relative to cost (Polyethylene). Full rankings are shown in Table 1. Material properties were entered in SI units ($E = 1.2 \times 10^9$ Pa, $\rho = 900$ kg/m³, $\nu = 0.42$, yield strength = 3.5×10^7 Pa) and converted internally by Workbench to IPS units.

Table 1: Candidate material properties and stiffness-per-cost ranking ($M = E / \rho / \text{price}$). *Selected material.

Material	E (GPa)	ρ (kg/m ³)	ν	σ_y (MPa)	Price (\$/kg)	$M \times 10^{-4}$
Polycarbonate	2.7	1200	0.42	70	\$3.65	6.16
Polyethylene	1.0	950	0.42	25	\$1.10	9.57
Polypropylene*	1.2	900	0.42	35	\$1.35	9.88
ABS	1.9	1030	0.42	29	\$2.15	8.58
Polystyrene	1.9	1045	0.42	27	\$2.20	8.26
Polyvinyl Chloride	1.5	1370	0.42	53	\$1.55	7.06

2.2 Baseline Finite Element Model

A solid rectangular half-model (4 in $\times$ 15 in) was constructed in DesignModeler as a 2D plane stress surface body. The plane stress thickness was defined in the Mechanical Details panel for the surface body, ensuring proper 2D formulation. Preliminary thicknesses of $t = 0.5$ in and $t = 1.5$ in both satisfied all constraints with large margin on the solid baseline, confirming topology optimization was justified. A starting thickness of $t = 0.25$ in was adopted; the simplified axial estimate $\delta = PL / (AE) = (150 \times 15) / (8 \times 0.25 \times 174,045) = 0.00648$ in confirms order-of-magnitude validity.

Applied boundary conditions: $u_y = 0$ along the top edge (u_x free); $u_x = 0$ along the symmetry plane (u_y free); 75 lbf downward distributed force along the bottom edge, representing half the 150 lbf total load. The reaction force was verified at 75.0 lbf, confirming global equilibrium. The 3D model used for topology and supplementary verification used the same thickness $t = 0.223$ in as the 2D formulation, ensuring a direct comparison. All primary validation results were obtained from the 2D plane stress model.

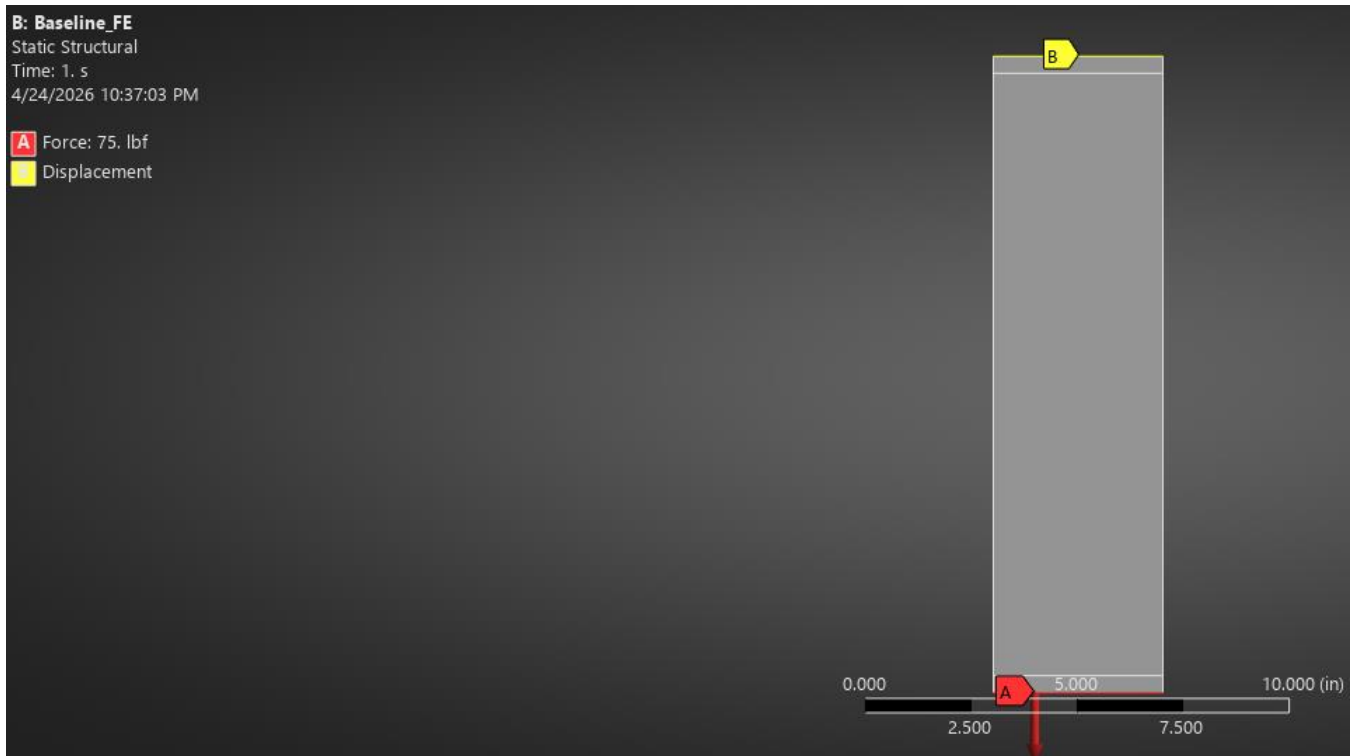


Figure 1. Boundary conditions applied to half-model: $u_y = 0$ (top edge), $u_x = 0$ (symmetry plane), 75 lbf distributed force (bottom edge).

2.3 Topology Optimization

Topology optimization was performed on a 3D solid body in ANSYS, as the topology module does not support 2D plane stress elements. The objective was minimize compliance (maximize stiffness) over the full solid domain, with no pre-seeded geometry. Non-design bands of 0.35 in were preserved at the top and bottom edges to prevent artifacts at load transfer interfaces.

Mass fraction evaluations from 25% to 55% showed: at 55%, coherent primary load paths form; at 40%, members refine further; at 25%, paths shrink significantly, indicating over-removal. The 55% result was used as the reconstruction reference. Topology output was interpreted qualitatively and not used directly as geometry, due to mesh dependency and lack of manufacturable continuity in the raw optimizer output.

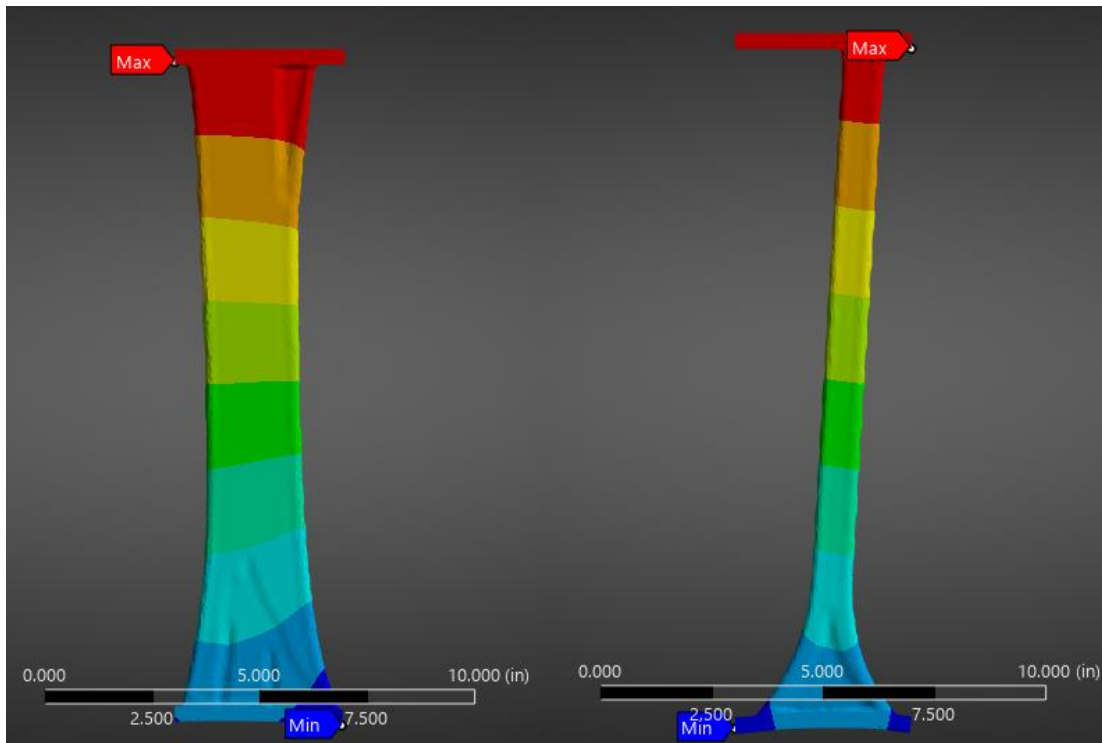


Figure 2. Topology optimization density distributions. Left: 55% retained mass, coherent load paths. Right: 25% retained mass, minimal support.

2.4 Geometric Reconstruction

Clean parametric geometry was sketched from scratch in DesignModeler using the 55% topology result as a visual reference, not imported directly. The smoothed topology output was first verified in a static structural run to confirm load path validity before the manual CAD rebuild. The final geometry is a symmetric truss-like planar profile with one triangular and two rectangular internal voids, a curved outer boundary, and fully filleted internal corners to eliminate stress singularities. Load path continuity from bottom to top edge is maintained with no isolated features.

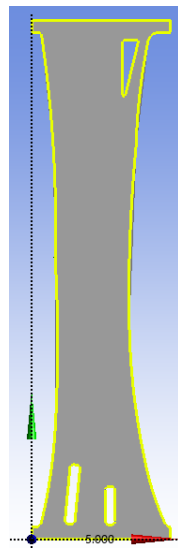


Figure 3. Reconstructed parametric geometry: tapered truss-like planar profile with one triangular and two rectangular internal voids, curved outer boundary, and filleted internal corners.

2.5 Parametric Optimization

Parametric optimization in ANSYS DesignXplorer minimized solid volume subject to max total deformation at or below 0.01475 in. Six input parameters were defined: global thickness t (0.20-0.30 in), outer edge fillet radius, triangular cutout fillet radius, small rectangular cutout fillet radius, large rectangular cutout fillet radius, and top and bottom bar thicknesses (all 0.20-0.35 in). Three algorithms were evaluated: MOGA provided broad global search; NLPQL offered rapid local convergence; ASO used a metamodel approach to achieve the lowest feasible volume across all 185 design points and was selected as the primary result. The optimal candidate (DP 44) yielded a half-model volume of 8.1347 in^3 and max deformation of 0.01478 in. The geometry was manually adjusted slightly to produce cleaner dimensions, giving a final half-model volume of 8.0311 in^3 and $t = 0.223 \text{ in}$. Cost is computed for the full bracket by doubling the half-model volume. Table 2 tracks volume evolution across all stages.

Table 2: Volume and mass evolution across optimization stages.

Stage	Half-Model Volume (in^3)	Mass (lbm)	Max Deformation (in)
Solid baseline ($t = 0.25 \text{ in}$)	15.000	0.4877	0.00646
Post-topology (55% retained)	9.1277	0.2968	0.01249
Parametric optimum (ASO, DP 44)	8.1347	0.2645	0.01478
Final design ($t = 0.223 \text{ in}$)	8.031	0.261	0.01492

A	B	C	D	E	F	G	H	I
Name	Parameter	Objective			Constraint			
		Type	Target	Tolerance	Type	Lower Bound	Upper Bound	Tolerance
Minimize P7	P7 - Solid Volume	Minimize	0		No Constraint			
P8 $\leq 0.01475 \text{ in}$	P8 - Total Deformation Maximum	No Objective			Values $\leq$ Upper Bound		0.01475	5E-05

Input Parameters			
Name	Lower Bound	Upper Bound	
P1 - Thickness (in)	0.2	0.3	
P2 - Outer_Edge_Fillet (in)	0.025	0.095	
P3 - Tringle_Fillet (in)	0.025	0.095	
P4 - Small_Rect_Fillet (in)	0.025	0.095	
P5 - Big_Rect_Fillet (in)	0.025	0.095	
P9 - Top_Bar_Thickness (in)	0.2	0.35	
P10 - Bottom_Bar_Thickness (in)	0.2	0.35	

Figure 4. DesignXplorer parametric setup showing six input parameters and the Adaptive Single-Objective optimal design point (DP 44): volume = 8.1347 in^3 , max deformation = 0.01478 in.

2.6 Mesh Strategy and Convergence Validation

The final geometry was meshed using triangular elements with three size controls: global element size, curvature-capture minimum, and local face sizing at fillets. The final reported mesh used global 0.150 in, curvature minimum 0.025 in, and face sizing 0.050 in, producing 119,438 nodes and max deformation 0.014922 in. The plane stress thickness was defined in the Mechanical Details panel for the surface body, ensuring proper 2D formulation.

Mesh independence was validated via a separate 47-point ASO study sampling global sizes 0.125-0.300 in, curvature minima 0.025-0.100 in, and face sizing 0.045-0.150 in, with node counts from 19,427 to 76,192. Max deformation ranged 0.014922-0.014951 in, a variance of 0.194%. The maximum change between successive refinements was below 0.3%, confirming mesh independence. The oscillation across the full node count range represents numerical noise below 0.2% tolerance, confirming numerical saturation. A representative subset is in Table 3; convergence plots are in Appendix Figure A5.

Table 3: Representative mesh convergence results (47-point ASO study). *Final design mesh. Full dataset in Workbench archive.

DP	Global Size (in)	Curv. Min (in)	Face Size (in)	Nodes	Max δ (in)
Final*	0.150	0.025	0.050	119,438	0.014922
1	0.205	0.066	0.102	35,772	0.014934
3	0.132	0.047	0.111	72,125	0.014924
9	0.191	0.028	0.084	76,192	0.014922
14	0.283	0.100	0.144	19,427	0.014941
17	0.191	0.100	0.061	49,031	0.014943
...	...	...	...	...	...
47 pts total	0.125-0.300	0.025-0.100	0.045-0.150	19,427-76,192	0.014922-0.014951

3. Results and Discussion

3.1 Summary of Final Design

Table 4 summarizes all final results. All figures are in the Appendix. The complete Workbench archive is submitted alongside this report.

Table 4: Final design results summary.

Parameter	Value
Material	Polypropylene (PP)
Thickness, t (in)	0.223
Full model volume (in ³)	16.0622
Mass (lbm / kg)	0.5223 lbm / 0.2369 kg
Material cost per part (\$)	\$0.320
Max $ u_y $ - 2D plane stress (primary) (in)	0.014772
Max $ u_y $ - 3D supplementary verification (in)	0.014739
Maximum total deformation - 3D (in)	0.014922
Maximum von Mises stress - 3D (psi)	332.02
Material yield strength (psi)	5,076
Factor of safety (von Mises)	15.3

3.2 Solid Model

Appendix Figures A3 and A4 show the final CAD geometry and the initial 3D verification result (max y-deflection = 0.01323 in), respectively, confirming feasibility prior to constraint saturation.

3.3 Finite Element Mesh

The final mesh is shown in Appendix Figure A6. Local refinement at fillets via curvature capture and face sizing produced 119,438 nodes. Deformation variance across the 47-point convergence study was 0.194%.

3.4 Von Mises Stress Distribution

Appendix Figure A8 presents the von Mises stress contour. The maximum stress of 332.02 psi occurs near the top constraint interface, yielding $\text{FoS} = 5,076 / 332.02 = 15.3$, well above the required 2.0. Localized concentrations at internal cutout corners reach approximately twice the main body stress level; these are expected geometric effects at re-entrant corners and remain far below yield. Fillets eliminate true singularities at all corners. The design is governed entirely by the deflection constraint.

3.5 Vertical Deflection Distribution

The primary deflection verification is the 2D plane stress result (Appendix Figure A9): maximum $|u_y| = 0.01477$ in, at 98.5% of the 0.015 in limit. The 3D supplementary result (Appendix Figure A10, same thickness $t = 0.223$ in) gives 0.01474 in, within 0.22% of the 2D value, confirming the plane stress assumption is valid. Maximum total deformation in 3D is 0.014922 in (Appendix Figure A11), also within limit.

3.6 Discussion

The design is entirely stiffness-limited: the deflection constraint operates at 98.5% capacity while the stress factor of safety (15.3) is seven times the required minimum. This confirms that the stiffness-per-unit-cost material selection strategy was the correct governing logic. Further material removal attempts violated the 0.015 in deflection constraint, confirming the design is near-optimal. Fillets at all internal corners eliminate singularities; the observed 2x stress concentration at cutout corners is a local effect and poses no yield risk.

Additional optimization avenues exist but were not pursued due to diminishing returns. A second topology pass on the reconstructed geometry could refine void placement. Shape optimization of the model could smooth stress gradients. optiSLang with its Metamodel of Optimal Prognosis (MOP) could survey the parameter space more efficiently than ASO. All are rendered marginal by the current design's limit.

4. Conclusion

A minimum-cost plastic bracket was designed and verified using a multi-stage FEA workflow in ANSYS Workbench 2025 R2. Polypropylene was selected based on the highest stiffness-per-unit-cost figure of merit ($M = 9.88 \times 10^{-4} \text{ GPa} \cdot \text{m}^3 / (\text{kg} \cdot \$)$) among six candidates.

The workflow proceeded through topology optimization at 55%, 40%, and 25% retained mass fractions, manual geometric reconstruction in DesignModeler, and 185-point parametric optimization using MOGA, NLPQL, and ASO algorithms. The ASO result was manually refined to $t = 0.223$ in and full-model volume of 16.0622 in³.

Both specifications are satisfied. Maximum y-deflection: 0.01477 in (2D, primary) and 0.01474 in (3D supplementary), both below the 0.015 in limit. Maximum von Mises stress: 332.02 psi, $\text{FoS} = 15.3$, exceeding the required 2.0. Reaction force verified at 75.0 lbf. Total cost: \$0.320, from full-bracket mass 0.2369 kg at \$1.35/kg. Mesh independence confirmed across 47 points with 0.194% deformation variance.

Appendix: Figures

All figures referenced in the main report body are provided below.



Figure A1. Boundary conditions applied to half-model: $u_y = 0$ (top edge), $u_x = 0$ (symmetry plane), 75 lbf distributed force (bottom edge).

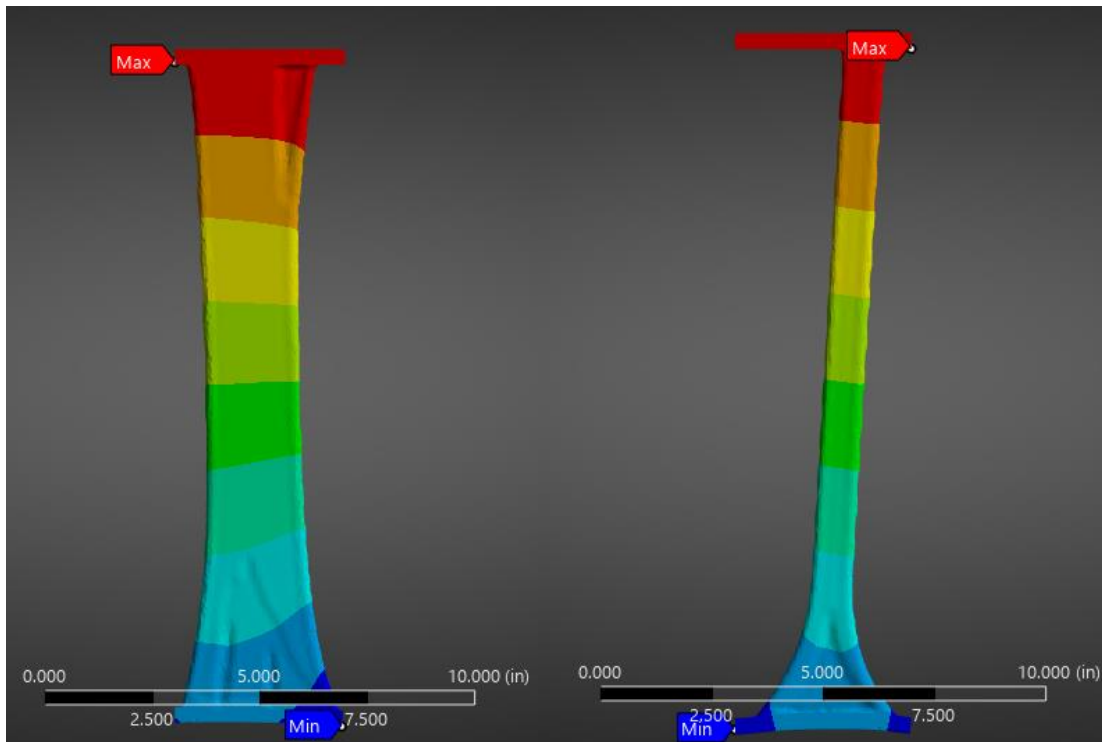


Figure A2. Topology optimization density distributions. Left: 55% retained mass, coherent load paths used for reconstruction. Right: 25% retained mass, fragmented beyond useful interpretation.

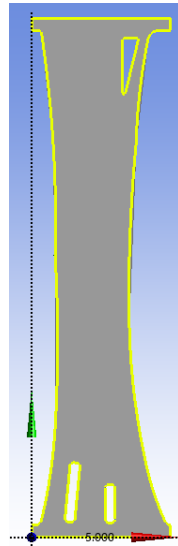


Figure A3. Reconstructed parametric geometry: tapered truss-like planar profile with one triangular and two rectangular internal voids, curved outer boundary, and filleted internal corners.

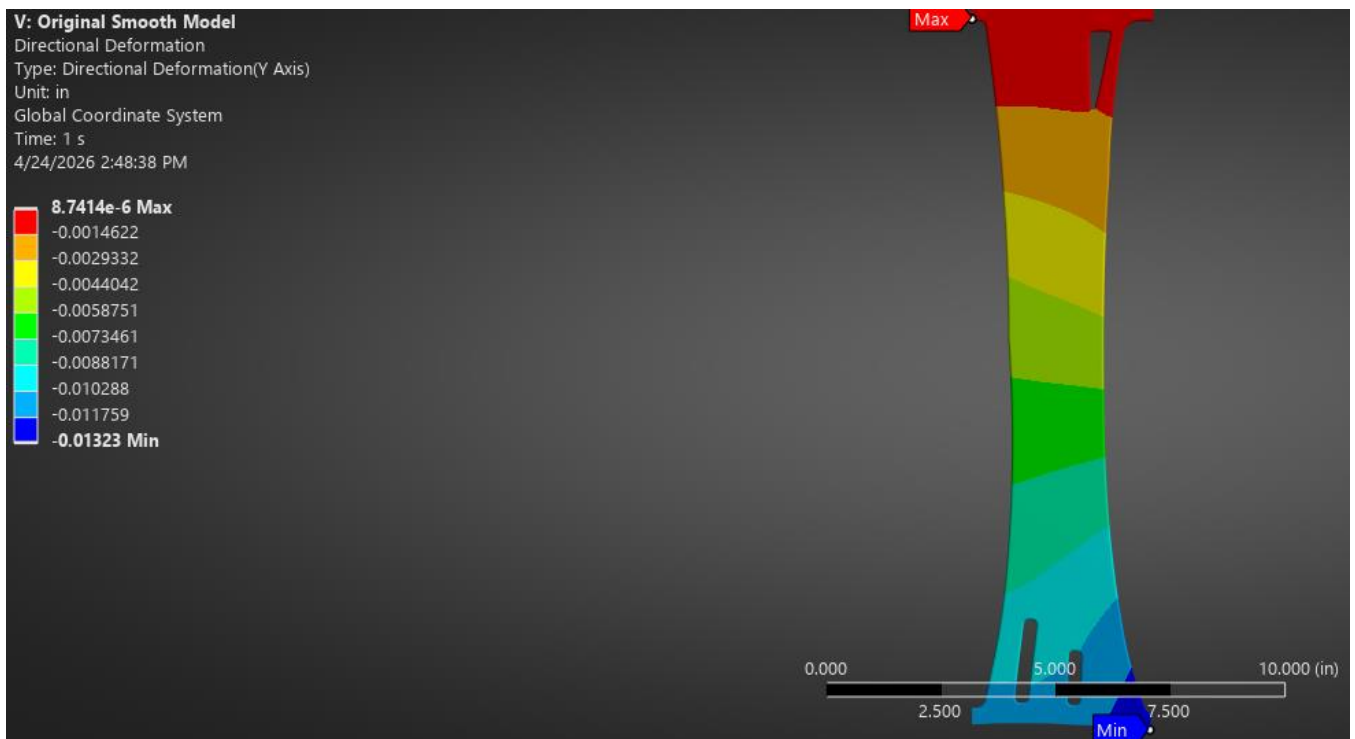


Figure A4. Initial reconstructed geometry verified in 3D static structural prior to parametric optimization. Max y-deflection = 0.01323 in, confirming feasibility before constraint saturation.

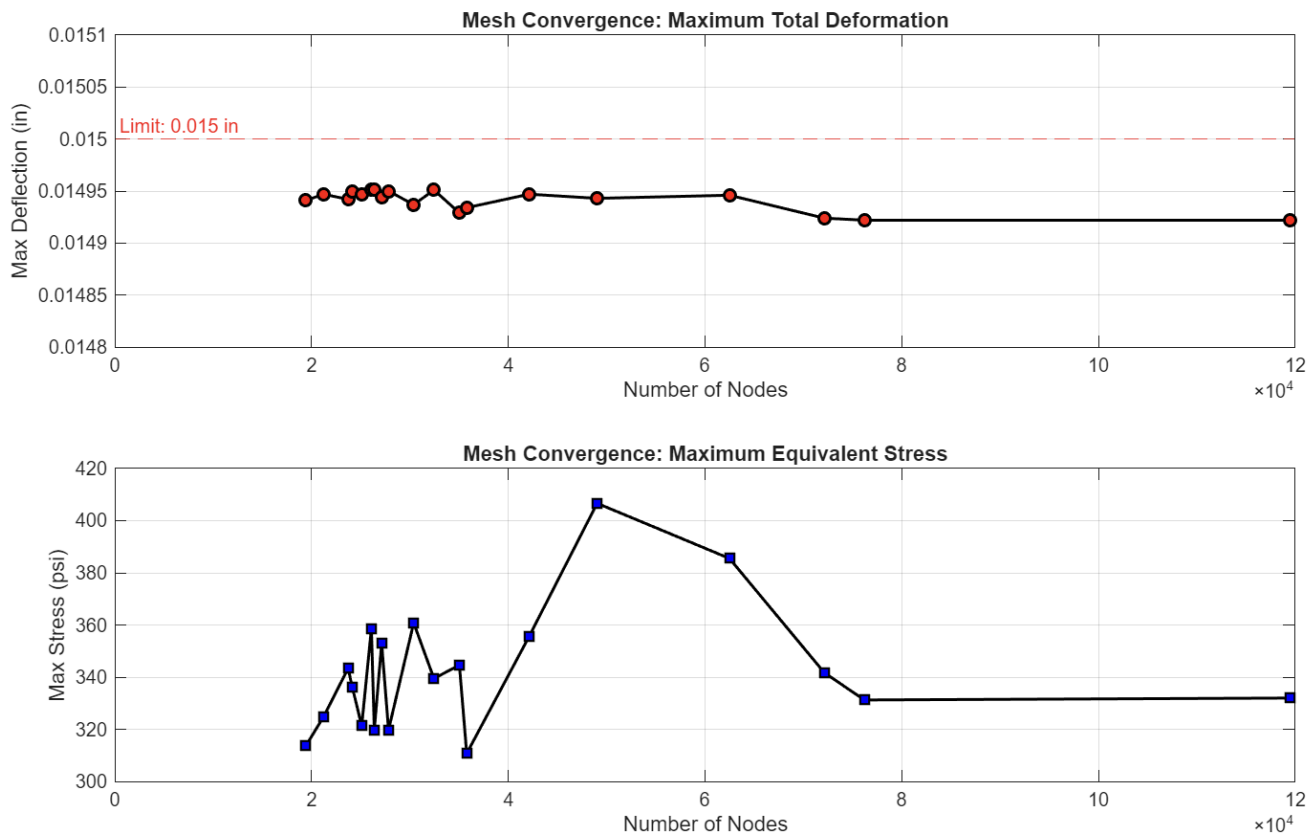


Figure A5. Mesh convergence study (47-point ASO). Maximum deformation variance: 0.194%. Oscillation between 20,000 and 76,000 nodes represents numerical noise below 0.2% tolerance, confirming numerical saturation. Stress range 311-407 psi, all well below yield (5,076 psi).

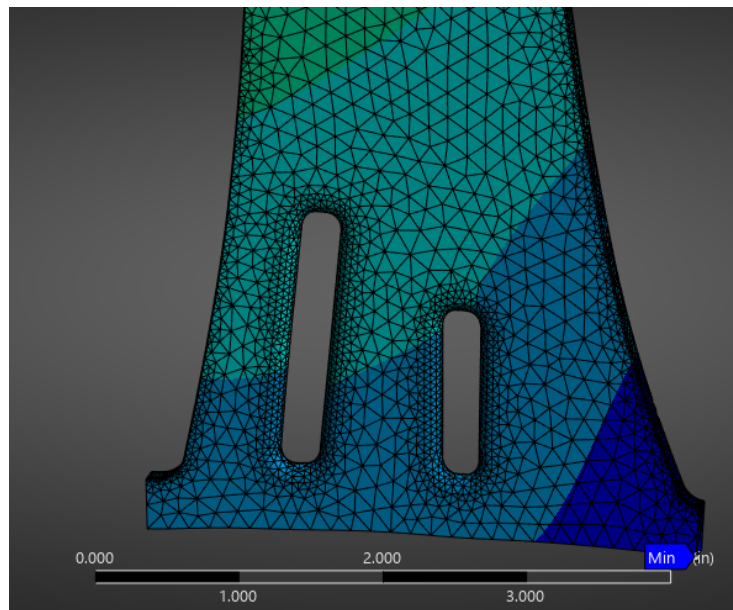


Figure A6. Final mesh: triangular elements, curvature capture min 0.025 in, face sizing min 0.050 in. Final design point: 119,438 nodes.

A	B	C	D	E	F	G	H	I
Name	Parameter	Objective			Constraint			
		Type	Target	Tolerance	Type	Lower Bound	Upper Bound	Tolerance
Minimize P7	P7 - Solid Volume	Minimize	0		No Constraint			
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P9 - Top_Bar_Thickness (in)	0.2	0.35	
P10 - Bottom_Bar_Thickness (in)	0.2	0.35	

Figure A7. DesignXplorer parametric setup: six input parameters, ASO algorithm, 185 design points. Optimal candidate DP44: volume = 8.1347 in³, max deformation = 0.01478 in.

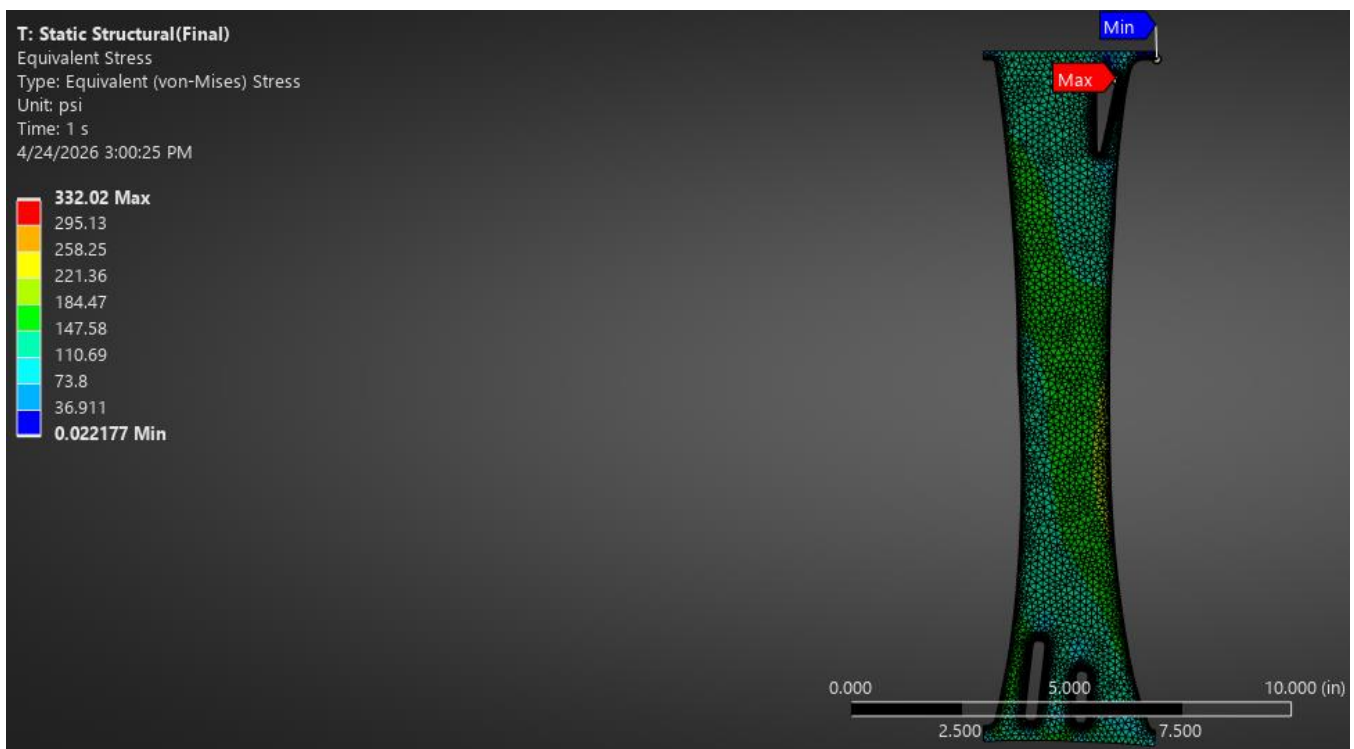


Figure A8. Von Mises equivalent stress contour under 150 lbf total load. Maximum stress = 332.02 psi at top constraint interface. Localized concentrations at internal cutout corners are approximately 2x main body stress and remain well below yield. Factor of safety = 15.3.



Figure A9. Y-direction deformation contour, 2D plane stress (primary verification). Maximum $|u_y| = 0.014772$ in. Allowable limit = 0.015 in. PASS.

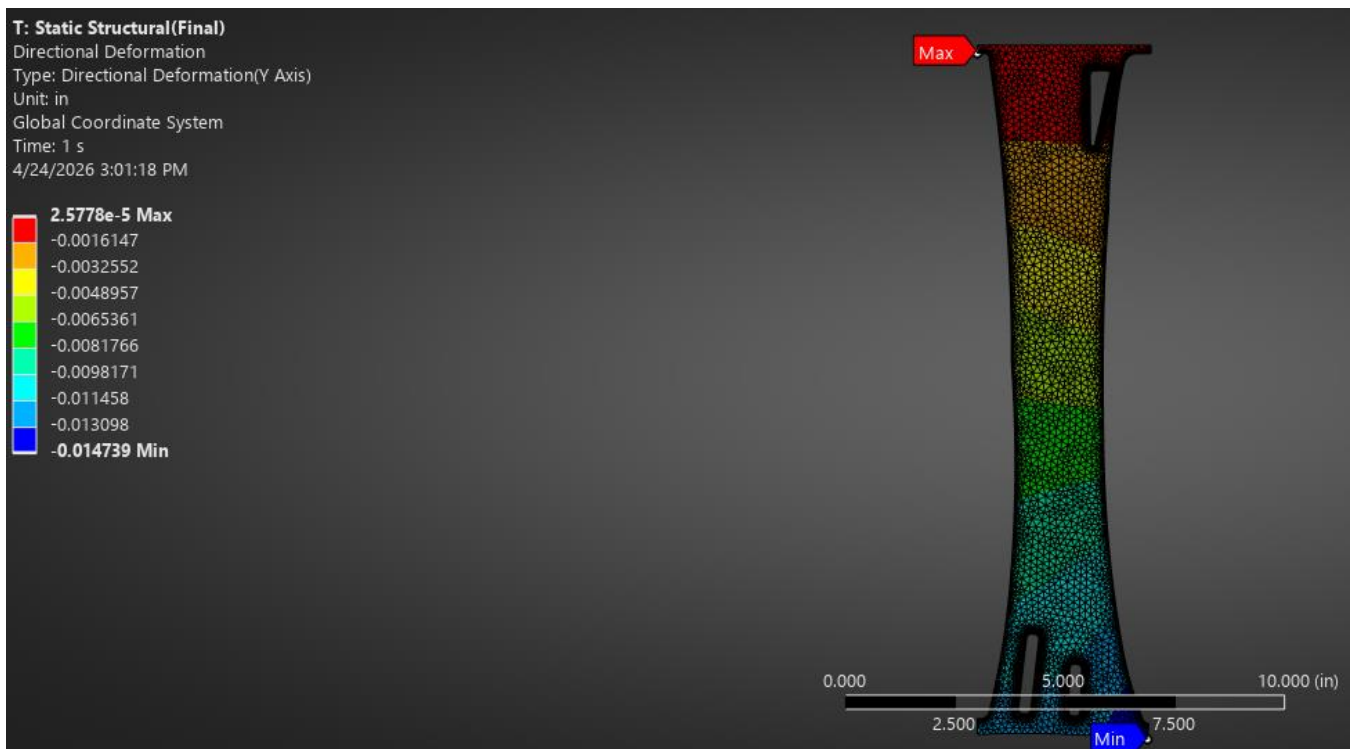


Figure A10. Y-direction deformation contour, 3D supplementary verification ($t = 0.223$ in). Maximum $|u_y| = 0.01474$ in. Agreement with 2D result within 0.22%.

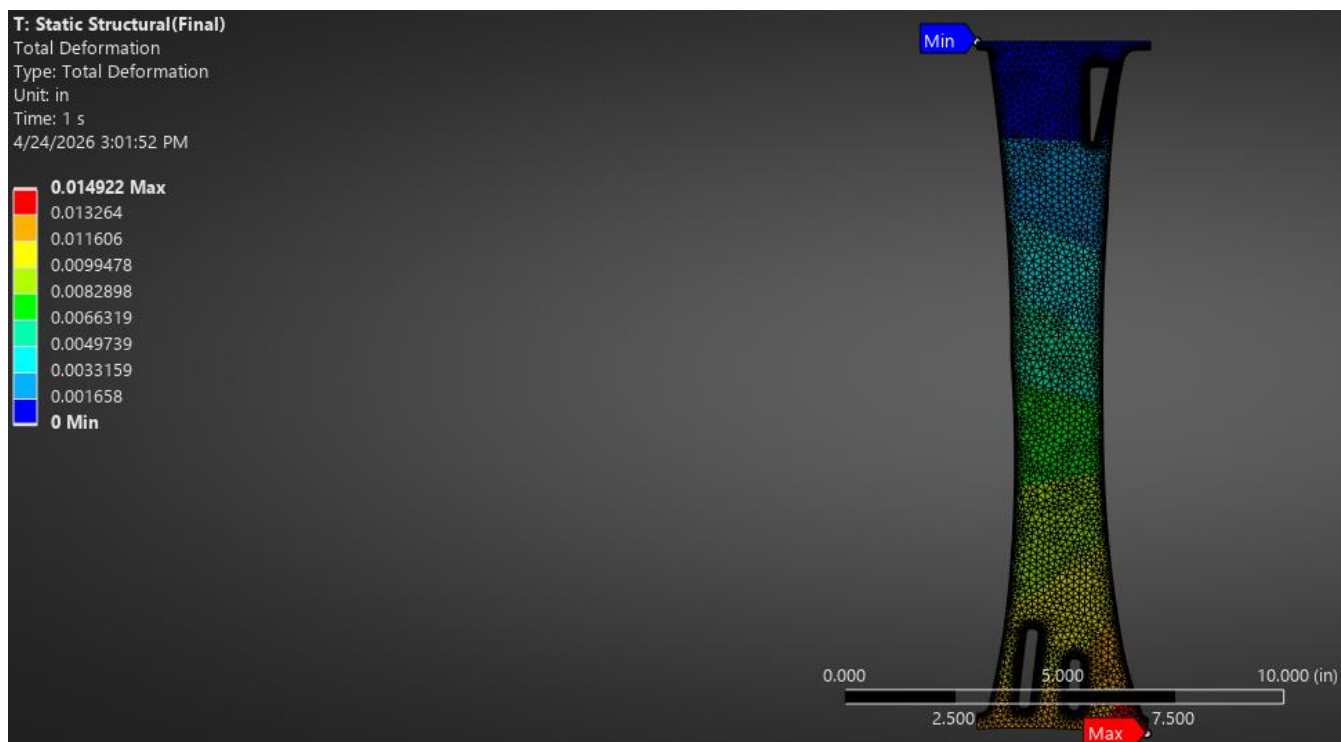


Figure A11. Total deformation contour, 3D supplementary verification. Maximum = 0.014922 in. All nodes within 0.015 in limit.